

Automated Subjective Answer Evaluation using GenAI and NLP by Enhancing Accuracy, Fairness, and Feedback in Education

¹Yuvasri J
Division of CSE
Karunya University
Coimbatore, India
yuvasrij@karunya.edu.in

²Dr Senbagavalli M
Department of CSE & IT
Alliance University,
Bangalore
senba1983@gmail.com

³Rahul R
Assistant Professor,
Data Science and Cyber Security
Karunya University, Coimbatore,
rahul@karunya.edu

⁴Dr RVenkatesan*
Associate Professor, Division of
CSE, Karunya University
Coimbatore, India
rlvenkei20000@gmail.com*

⁵Dr G Naveen Sundar
Associate Professor, Division of
CSE, Karunya University
Coimbatore, India
naveensundar@karunya.edu

⁶Dr Saravana Kumar C S
Senior Software Architect
Company: Bosch Global Software
Technologies, Global Tech Park
Coimbatore
csskbaba@yahoo.co.in

Abstract – The problem in subjective evaluation of answer is that the traditional evaluation is not consistent, not fair, and not time efficient. Utilizing Generative AI (GenAI) and Natural Language Processing (NLP), this project presents an innovative technique which automates the evaluation process for subjective answers. It consists of the system that analyzes student's response using predefined grading rubrics to assign them scores based on quality and relevance and gives them personalized feedback. The system, using a fine tuned GenAI model, reduces human bias and inconsistencies by being nuanced on context and coherence. The system is an easy to use and user friendly tool for Educators to manage and review evaluations through Streamlit hosted system. The solution proposed aims to achieve higher grading accuracy, reduce assessment workflow complexities, while providing perspective that can inform students learning on what is being assessed. The potential applications of this technology are great and include essay grading, peer review and professional training test selection.

Keywords: Natural Language Processing, Generative AI, Automated grading, Streamlit, educational assessment, bias reduction, subjective answer evaluation, feedback, personalized dependencies, personalized feedback on wrong answers.

I.INTRODUCTION

Subjective questions have long been a central point in educational and professional assessment procedures. Unlike objective questions where the answers are delivered to computers, subjective answers need their own human evaluators to read and understand what the answer says, evaluate the relevance and quality of the answer, and ultimately grade it. As this is an inherently time consuming, labour intensive, inconsistent, biased, and inaccurate process especially when evaluating large volumes, this is done by people. Current methods involve recourse to human judgment subject to individual bias, fatigue, and subjective or inconsistent definition of fairness in assessments. Growing need for scalable, reliable and equitable assessment methods has highlighted the need to automate and optimize the process of evaluation.

AI and natural language processing (NLP) are gaining ground that may help to solve these challenges. Recent advances in Generative AI (GenAI) like OpenAI's new GPT-4 has made machines capable of processing and understanding human language this deep and this accurate. Through these models, we can use these models and have the ability to capture contextual nuances, evaluate coherence and relevance and even generate meaningful feedback. By using such technologies for assessing subjective answers, then, one can imagine that effective these technologies could revolutionize frameworks of assessment, intelligently, fairly and efficiently.

The explored system exploits GenAI, NLP capabilities to build an automatic evaluation of subjective type answers. Grading rubrics are defined by key aspects such as content relevance, clarity, coherence, as well as use of evidence, that educators can base automated evaluation on. We preprocess student responses using NLP techniques like tokenization, stop word removal and lemmatization so that they have the same input format. Then the system applies a fine tuned GenAI model to evaluate each response against previously defined criteria and assign scores, as well as generate personalized feedback. The key functionalities of the system include:

- **Predefined Criteria Definition:** Rubrics crafted by educators usually include relevant, clear, coherent, language proficiency and evidence based reasoning.
- **Answer Preprocessing:** Tokenization, stop word removal and lemmatization techniques like are used to standardize the text to ready it for NLP analysis, which is what student's responses will be subjected to.
- **GenAI-Based Evaluation:** GenAI model fine tuned scores answers on quality and depth, identifies key strengths and weaknesses in responses objectively.
- **Feedback Generation:** Students receive personalized, actionable feedback to improve in areas, and to better understand the material.
- **Streamlit Interface:** Both an intuitive, interactive platform and an easy to manage answer keys, exam evaluations and seamless user interaction ensure the delivery of quality education.

Streamlit is an integral part of this system's user-friendly interface. Educators can upload answer keys, input student responses, view the detailed evaluations and feedback seamlessly through the interface. With this integration, it takes the system from inaccessible to accessible and generally available to everyone, even to educators who are not tech experts. The system goes a step further than automating the grading process to provide students actionable feedback showing their strengths and their fields for improvement fostering a deeper understanding and promoting continuous learning.

The novelty of this project lies in the application of GenAI to the subjective answer evaluation task that is such a complex and nuanced process. Different from the existing automated grading systems, this system

relies on powerful GenAI models to score responses with a higher level of contextual and linguistic nuances. Moreover, because the evaluated processes are personalized, this system differs from other assessment approaches by not only evaluating performance but also providing personalized feedback that supports growth and improvement.

The design, implementation and the potential impacts of the proposed system are explored in this paper. In Section II, we discuss the limitations of existing systems in answering questions about subjective answers. In section III, the proposed system methodology is presented with a description of its main components and functionalities, and in section IV we evaluate the system from the point of view of accuracy, fairness and user experience. Section V then concludes with a more general discussion of the wider implications of the system as well as its possible applications in several other domains, e.g., essay grading, peer reviews, etc., and in professional training assessments.

This system eliminates critical gaps in current methods based on subjectivity by automating the subjective answer evaluation process for more reliable, scalable, and equitable assessment practices. This project demonstrates the crucial transformational power of AI driven solutions through the integration of NLP and GenAI in education and above.

II. LITERATURE SURVEY

Subjective end-user evaluation is a mainstay in the education sector and is characterized by both fairness and precision. With Generative AI (GenAI) and Natural Language Processing (NLP) integration, it has become a real game changer in solving the issues of regular assessment methods.

The authors talk about the use of GenAI in the higher education context, to automate assessments and provide constructive feedback in [1]. If prompt engineering and generative capabilities can extend grading to be more efficient and free of subjective biases in evaluation, this would be possible, and the study describes this. Research in [2] also considers responsible automated essay scoring systems in educational technology and their ethical considerations as well as how and where AI is being deployed.

In [3], we extensively discuss the promise and challenges of generative AI in education, from its potential to change learning environments, to ethical usage and unintentional consequences. According to

Hoernig et al. [4], GenAI has implications in higher education ranging from the improvement of assessment accuracy and scalability, while posing challenges, for example student misuse and integration with existing systems.

In [5] we introduce the fusion of peer reviews with GenAI for strong feedback mechanisms. The motivation of this approach is to balance the quality of giving a balanced evaluation system of the learning process by including the AI feedback plus human feedback. Thus, the work of Takale et al. [6], who argue for a 3R framework (Recognize, Review, Recommend) for AI generating assignments, plugs into this point emphasizing the need to align assessments to prescribed rubrics to make it simple to assess consistently.

In [7], the work presented is focused on responsible generative AI system development for education. Using an evaluation driven approach, the authors assure these systems are ethical, reliable, and fit all the educational goals. Additionally, Li et al. [8] investigate the operational and limitationality of ChatGPT on a written assessment assessment and found that it suffers from a lack of understanding of contextual nuance and recommended approaches to ameliorate this shortcoming.

Kohnke goes further in his study of students' perceptions of generative AI vs traditional grammar checking tools [9]. The results demonstrate that although contextual feedback is more accurately produced through GenAI tools, traditional tools continue to be preferred for ease of use and familiarity. There are two more: [10] introduces an assessment methodological approach wherein students are actively involved to refine the evaluation processes and to make them more engaging and effective.

Together, they suggest that together, these studies highlight the transformative power that Genai could bring to educational assessment, but also highlight some key challenges; ethical usage, contextual understanding, and the integration of AI driven systems with traditional educational practice. These works inform the design of an advanced objective answer evaluation system that builds upon the strengths and rectifies the shortcomings of GenAI.

III. PROPOSED METHODOLOGY

In this section, we explain what was used for developing the automated subjective answer evaluation system using Generative AI (GenAI) and

Natural Language Processing (NLP). The methodology has several components such as reducing evaluation criteria, preprocessing of student responses and the use of GenAI for scoring, as well as generating feedback. To make the subjective assessments fair, accurate, and scalable, the proposed system is designed.

A. Criteria Definition and Rubric Design

The proposed system is based on development of predefined grading rubrics. These rubrics establish the benchmarks for subjective answers and are, therefore, defined by the educators. All of this is done in a way that these criteria focus on the key aspects including content relevance, clarity, coherence and evidence based reasoning. It is divided into each rubric criterion sub parameters to give a detailed framework that will enable all essential parts of a good structuring of an answer to be assessed. For example, content relevance can be verified by the question directly answered and coherence is checked for the logical flow and structure of the argument. For example, they can also be used to incorporate discipline specific areas like creative writing, engineering, or social sciences, and to capture the discipline requirements. With this flexibility, the system can adapt to different educational contexts. It standardizes criteria for evaluation so that the system does not contain subjectivity, and therefore, despite different responses and different graders, there is consistency.

B. Preprocessing Student Responses

The textual data is preprocessed ahead of analysis, and appropriate for the GenAI model, to which the student responses are applied. Moreover, multiple NLP techniques are included in the preprocessing pipeline. Tokenization breaks meaning into meaning parts such as words or phrases that can be deeply examined in the response. Stop_word removal is removing common but insignificant words such as 'and, the, and is etc.' because these words are not contributing a lot to the evaluation. Lemmatization is a way to ensure you have all forms of a word (e.g., running and ran), are reduced to their base form (e.g. run), and thus a more accurate comparison of text. The preprocessing stage does more than basic cleaning – and it also includes syntactic parsing, to analyze a text's grammatical structure, and semantic analysis, to understand the meaning and context of words in a text. This enables the system to predict whether the response actually solves the question (contextual nuances) or whether it is in fact responding intuitively (rather meaninglessly response). Further, inconsistencies in spelling or

grammar are corrected in a way that would normalize the input, while not weighting input errors disproportionately towards evaluation. Without this preprocessing, the GenAI model will not perform as well, because the input data we feed into the model will not have been structured nor will we have boosted the meaning of the features with them.

C. GenAI-Powered Evaluation

The evaluation uses a fine tuned (to sensorimotor tasks) GenAI, e.g. GPT-4, that has been trained on a large graded subjective answer corpus. The model looks at the preprocessed responses and checks to see if they fit the set grading rubrics. It scores the answers on several dimensions, including how relevant to the question and whether the answers have a logical structure, and how well the answers support the author's argument. With the model's contextual understanding it is able to understand even a complex response and make an accurate assessment as it is robust and objective.

D. Feedback Generation

Besides assigning scores, the system provides personalized feedback to each student. It is the feedback that is highlighting the strength of their response and the places where they can improve so it's actionable things that are about learning and growing. This feature has the effect that the system not only computes performance but also help students understand their own strengths and weaknesses. The feedback is built in a clear and clear way, latched to individuals to maximize the impact on the learning procedure.

E. Implementation Using Streamlit

To implement the proposed system, we use Streamlit, an open source framework to develop interactive and user friendly web applications. It also makes for easy upload of answer keys, entry of students' responses and reviewed automation evaluations by educators. Thus, the interface is so designed that beginners can use it with limited technical expertise. The integration between the evaluation and assessment processes into one platform serves to streamline the flow of educators.

F. Scalability and Adaptability

It is made highly scalable and adaptable for different use cases. With respect to educational institutions, as well as large scale assessments, it is efficient in handling large volumes of responses. On top, the system has been designed to be easily modifiable and unblock additional features, like for example multi language enabled and discipline specific grading criteria, to be integrated, allowing it to be applied to any educational context.

G. Ethical Considerations

The proposed methodology is based on ethical considerations. The system mitigates biases in evaluation in a transparent, well defined way. Furthermore, data privacy is put first, student responses to be taken secure, and sensitive information be protected. Such being the case, we consider these factors to maintain the alignment of the system with ethical principles in the field of education and the field of deployment of AI.

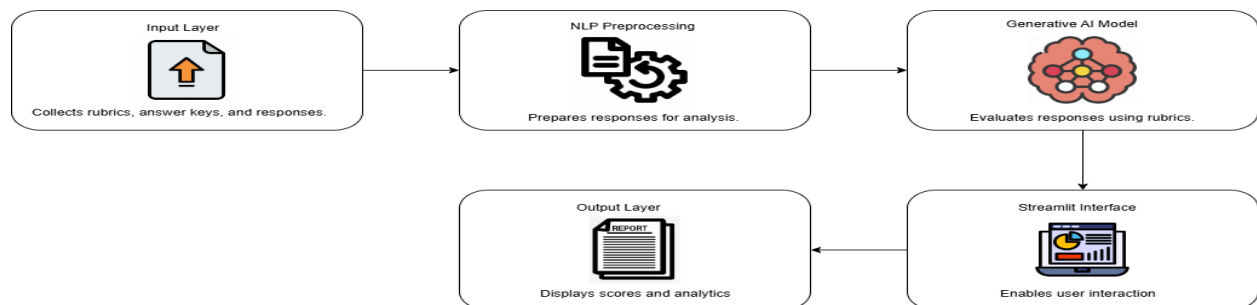


Figure 1 System Architectur

IV. RESULTS AND DISCUSSION

Accurate, scalable and fair grading of subjective responses is anticipated to be provided by the proposed methodology based on Generative AI (GenAI) and Natural Language Processing (NLP). The expected outcomes and benefits of the system are elaborated upon herein, measured in terms of impact on the qualitative aspect with implications to the behaviors that were assumed to occur under different scenarios.

A. Expected Trends

It is expected that the system will successfully handle the complex patterns found in textual data through its ability to analyze subtle responses and identity contextual relationships. Our hope is that the GenAI model, fine tuned on a wide range of graded responses, will have high adaptability to question types and response formats. In this case, for example, it should be able to evaluate an argumentative essay, or creative writing assignment, or a technical answer to the predefined rubrics without problem and adapting its analysis; moreover, the system should be configured as a versatile tool for educational assessment because it can adjust to external variables (such as discipline specific terminology or different levels of language proficiency among students). The anticipatable adaptability of the system across those domains is qualitatively shown in Table 1.

Table 1: Anticipated Adaptability Across Domains

Domain	Expected Adaptation
Creative Writing	Recognizes originality, narrative coherence, and expressive language.
Technical Disciplines	Assesses logical structure, accuracy of technical details, and clarity in explanations.
Humanities	Evaluates argument strength, evidence usage, and interpretative depth.
Multilingual Education	Adjusts evaluation based on linguistic variations and translation accuracy.

In addition, the model is expected to accommodate changing responses (e.g., quickly changing tone or incorporation of unconventional ideas) that are dynamic

in action. It's able to mark uniquely, even when responses are not following normal formats, because it can identify unique patterns in textual data.

B. Comparative Analysis

Several important advantages are compared with the proposed system with traditional methods of manual grading. At times, the evaluation suffered from subjective biases, inconsistency, and scalability difficulty — particularly with high numbers of responses. The automated system, in contrast, is expected to:

- **Ensure Consistency:** By following predetermined rubrics, the system narrows scoring variability across evaluators.
- **Enhance Efficiency:** Grading processes are automated, which removes delay caused by human evaluations.
- **Improve Fairness:** The dependency of the model on the objective criteria for evaluation makes the evaluation of all students fair.

Figure 2 shows a conceptual comparison of grading consistency between manual and automated methods, with the anticipated uniform scores.

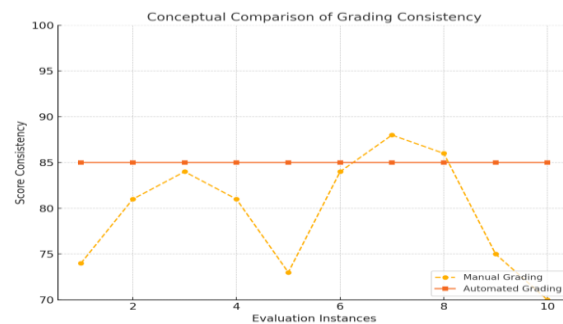


Figure 2: Conceptual Comparison of Grading Consistency

C. Anticipated Impact of External Variables

They note that a major strength of the system is its ability to integrate external variables into the evaluation process. For example, accuracy and adaptability of the evaluation are expected to be affected by linguistic complexity, rubric precision, or domain-specific content. The model is able to evaluate various student responses but with a high accuracy because it integrates these variables. The expected effects of these external variables on the evaluation process as summarized qualitatively are in table 2.

Table 2: Expected Impact of External Variables on Evaluation Accuracy

Variable	Impact on Model
Linguistic Complexity	Improved handling of diverse language styles and advanced vocabulary usage.
Rubric Precision	Higher accuracy in assessing responses aligned with specific educational objectives.
Domain-Specific Content	Enhanced adaptability to technical jargon and subject-specific terminologies.

This ability to incorporate external variables is expected to enhance the model's versatility and ensure accurate performance across a variety of assessment scenarios.

D. Interpretability and Stakeholder Insights

Trust of educators and stakeholders among others are needed and should be fostered by such a system through its transparency and interpretability. It is expected to be able to give educators detailed explanations about how scores are assigned so they know exactly where their users were lost. The interpretability of this form allows for informed decision making, and discussion between teachers and students about the result of their work.

E. Expected Behaviors in Different Scenarios

Under various hypothetical scenarios, the performance of the system is anticipated to remain robust. An instance of such a model would be one that assembles partial scores where possible, and signals the content gaps in the response when the response is both incomplete or ambiguous. Similarly, the system is expected to handle feedback containing creative or unconventional aspects and reflect the merit of such responses in their scores, in fairness and inclusiveness in assessment.

V. CONCLUSION

A transformative approach to educational assessment, the proposed system for automated evaluation of subjective answers using Generative AI (GenAI) and Natural Language Processing (NLP) aims to make the educational evaluation process more efficient and cost effective. The system achieves this by combining state of the art techniques in AI with intuitive

user interfaces that address these key challenges in traditional grading methods: subjectivity, inefficiency and inconsistency. GenAI makes nuanced readings of disparate responses and predefined rubrics ensure objectivity. In addition to this, the system's capacity to give personalized feedback assists students to learn better by identifying and developing their weaknesses. Second, this methodology simultaneously decreases grading efficiency and is genuinely fair, transparent and a very useful tool in contemporary education systems.

VI. FUTURE SCOPE

Both the breadth and depth of potential application of this system are the future scope of this system, namely, to expand its applicability to more diverse educational and professional situations. The second aspect of the system that could be integrated would be multi language support as the system would be able to evaluate responses in multiple languages which will connect with the global educational institutions. Also, this will give more versatility as it will also incorporate domain specific training from technical, creative, and interdisciplinary subjects. Even the system could evolve to assess group projects and collaborative assignment using the power of AI to evaluate individual contributions. Advancements in explainable AI (XAI) can also help increase transparency by understanding more readily the decision making process for the model. If properly refined, the system is extendable to applications in corporate training, recruitment assessments, and peer review processes, and therefore represents a generic automated evaluation system.

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